

LDP separation - Proof of concept

Testing Document

Version 1

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General description and purpose

Purpose of test

The findings of test 1 “[Sink-float separation - Proof of concept](#)” have demonstrated the potential of high purities (> 99.9 %) of the two resulting fractions: low-density polymers (LDP) – PP, LDPE, HDPE – and high-density polymers (HDP) – PS, PET, PVC. In the subsequent steps, these fractions must be divided further into its components. This test investigates the next step for the separation of the LDP types, conducted in a similar proof-of-concept fashion (low fidelity, small scale).

Due to similar densities, a separation based on that material property is difficult (PP ~ 0.900 g/cm³, LDPE ~ 0.924 g/cm³, HDPE ~ 0.955 g/cm³ (Bearth, 2025)). Instead, the difference in the parts shape could be a useful distinguishing feature. As thin-walled plastic waste, like plastic bags or packaging, is often composed of LDPE (Shen & Worrell, 2014), its large surface area in relation to its mass could be utilized, making it susceptible to air streams. This effect can be tested by vacuuming.

Description of Test

For this test, the color coded plastic flakes from test 1 are used. There, twelve mixed plastic test batches were created with the following composition: PP 0.6 g, LDPE 0.2 g, HDPE 2.0 g, PS 2.4 g, PET 2.7 g, PVC 1.7 g (9.6 g in total). After the completion of the test, the twelve batches were combined into one and kept for later use.

This mixed plastic is now used as the test batch. It is spread out on a clean surface. Then, a vacuum cleaner equipped with a strainer at the inlet is moved over the spread-out plastic waste to extract the thin foils from the thicker pieces. For consistency, the plastic flakes pulled in by the air stream is called float fraction, while the fraction remaining on the surface is considered the sink fraction. By visually sorting the correctly and incorrectly separated flakes of the sink and the float fraction and measuring the resulting fractions’ masses, the purity of the separation can be calculated.

Calculations

Plastic separation results:

$$\text{Purity } \varphi_{float}: \quad \varphi_{float} = m_{float_{LDPE}} / m_{float}$$

$$\text{Purity } \varphi_{sink}: \quad \varphi_{sink} = m_{sink_{not\ LDPE}} / m_{sink}$$

$$\text{Purity } \varphi_{combined}: \quad \varphi_{combined} = (\varphi_{sink} + \varphi_{float}) / 2$$

Schedule of test with all steps required

1. Attach a strainer to the inlet of the vacuum cleaner (concave facing outward).
2. Test the existence of airflow through the strainer with your hand. (lowest intensity)
3. Spread **30 g** of the mixed plastic flakes on a clean surface. (Test **a1**)
4. Set up camera to record the separation behavior over the test timeframe and start recording.
5. Move the vacuum cleaner with the strainer over the mixed plastic while keeping a distance of **35 mm**.
6. Collect the aspirated fraction in a separate container.
7. Sort the float and sink fractions by color and measure the masses.

8. Mix the fractions well.
9. Repeat steps 3-7. (Test a2)

Risk analysis

Potential risks associated with this test

None.

Mitigation steps

None.

Risks after taking mitigation steps

None.

Tools and devices required

Tool	Check-in tick
1x laptop	X
1x camera (phone)	X
1x scale	X
Remaining mixed plastic of test 1	X
1x container	X
1x strainer	X
Adhesive tape	X
1x tweezers	X
1x calculator	X
1x flat surface	X
1x vacuum cleaner	X

Testing Results

All recordings of the separation process and pictures of the results are available [here](#).

Strainer prep and vacuum test:

Very high airflow even at minimum suction

Mixed plastic ratio from test 1:

PP	LDPE	HDPE
0.6	0.2	2.0
PS	PET	PVC
2.4	2.7	1.7

Separation:

Test	Float-output masses [g]				Sink-output masses [g]	
a1	PP	LDPE	HDPE		LDPE	
	0	0.492	0		0.270	
	PS	PET	PVC		Others	
	0.075	0	0		29.1	
a2	PP	LDPE	HDPE		LDPE	
	0	0.751	0		0.001	
	PS	PET	PVC		Others	
	0.049	0.074	0		29.06	

Note: for a2, the strainer was fitted with a solder tin wire with a length of 35 mm to ensure a constant distance to the surface and induce movement within the flakes.

Any unexpected behaviors:

- Very high specificity (almost only foils picked up)
- Fast clogging
- Foils must be on top; if underneath other plastic flakes, it does not move
- Adhesive residual on plastic pieces impedes the separation possibility
- a1: only 3 flakes of PS in float fraction
- a2: when flakes are moved slightly (with tip of solder tin wire) + constant distance to surface kept by wire almost all LDPE was collected.
- a2: especially for last few suctions, relatively more PET and PS was picked up
→ high purity of LDPE foil OR high purity of sink fraction
- a2: only one flake of LDPE in sink fraction, three flakes of PS and three flakes of PET in float fraction



Figure 1: Resulting fractions of run a2; correctly classified flakes in black containers, misclassified flakes in red containers.

Evaluation

Generally, this method is well suited to extract foil flakes (see Table 1). However, due to the low mass of the plastic foil, the purity of the sink fraction is not a good measurement. Therefore, the number of misclassified flakes were counted as well.

Both runs exhibited a high specificity towards collecting foil flakes. In the first run, many LDPE flakes were covered by particles of other material which resulted in the inability to be picked out by the vacuum. At the same time, only three PS flakes were aspirated. For the second run, a solder wire with a length of 35 mm was attached to the end. This allowed a constant distance from the surface and induced motion between the flakes. Here, the number of correctly aspirated flakes was much higher with only one being left in the sink fraction. While the picked-up flakes were almost purely LDPE in the beginning of the separation procedure, the further the extraction got towards no LDPE in the sink portion, the more PS and PET flakes were picked up as well. This is an indication that the targets of high purity for the floating fraction and the sinking fraction are conflicting.

Since the sink fraction has to undergo an additional step after this procedure and their recycling capabilities are substantially larger than for LDPE, targeting maximized purity for the sink fraction could be beneficial for a prototype.

Important for this separation method is a constant distance to the flakes (constant air stream), motion within the flakes to loosen the LDPE flakes that are stuck underneath another material, and adhesives from prior use of the plastic being removed beforehand.

Table 1: Purity results and number of misclassified flakes, sorted by combined purity.

Test	Purity float	Purity sink	Purity combined	Misclassified flakes in float fraction	Misclassified flakes in sink fraction
a2	85.93%	100.00%	92.96%	PS: 3, PET: 3	LDPE: 1
a1	86.77%	99.08%	92.93%	PS: 3	LDPE: >20

Bibliography

Bearth, R. (2025). *Plastic separation—A review* [Report]. ETH Zurich. <https://doi.org/10.3929/ethz-b-000732181>

Shen, L., & Worrell, E. (2014). Plastic Recycling. In E. Worrell & M. A. Reuter (Eds.), *Handbook of Recycling* (pp. 179–190). Elsevier. <https://doi.org/10.1016/B978-0-12-396459-5.00013-1>